

Optimization of Adder Tree Architectures

1. Overview of Adder Tree Optimization

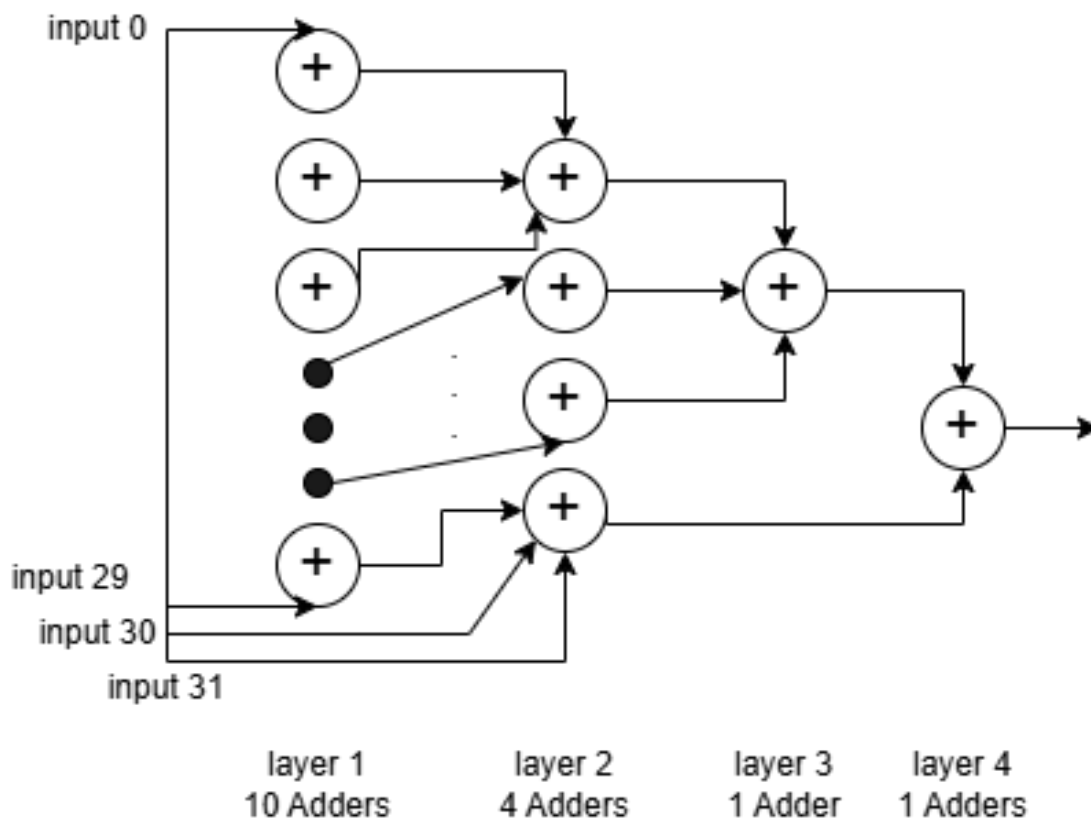
The primary objective of this architectural refinement is to minimize hardware resource utilization—specifically Digital Signal Processor (DSP) slices—by transitioning from traditional 2-input adder hierarchies to optimized 3-input structures.

2. Stage 2: 32-Input Adder Tree

In a standard binary adder tree, summing 32 inputs requires five layers of 2-input adders, totaling 31 units (arranged in layers of 16, 8, 4, 2, and 1).

16 → 8 → 4 → 2 → 1

By utilizing DSP slices that feature 3-input adders, the required number of DSPs is significantly reduced as shown in the following architecture:



Implementation Details:

- Layer 1: The first 30 inputs are processed by 10 3-input adders.
- Layer 2: The remaining 2 inputs of the original 32 are delayed and integrated into Layer 2, which consists of 4 adders.
- Resource Efficiency: Through this recursive consolidation across subsequent layers, the total hardware requirement is reduced to 16 adders, nearly a 50% reduction from the original 31-adder requirement.

3. Stage 3: 64-Input Adder Tree and Resource Sharing

A conventional 64-input adder tree typically requires 63 2-input adders across six layers (32, 16, 8, 4, 2, and 1).



To achieve greater efficiency and source sharing, the design utilizes a modular approach based on the Stage 2 architecture.

Optimization and Unified Architecture:

- Resource Scaling: By utilizing 6 repeated adder tree modules from Stage 2 to serve 192 inputs, the total number of adders is reduced to 33, compared to the 63 required by standard methods.
- Hardware Reuse: The architecture is designed to support both Stage 2 and Stage 3 operations within the same block by tapping the output at different points, thereby eliminating the need for complex multiplexers.
- Output Tapping:
 - Stage 2 Output: Extracted from Wire 1 and Wire 2 following the Layer 4 logic.
 - Stage 3 Output: Extracted from Wire 3 after the Layer 5 adder.

This configuration ensures that Layer 5 is only utilized during Stage 3 processing, maintaining a compact and efficient hardware footprint.

